

Modelor: An Early 2D Resource Modeling Tool for Tabular Deposits

Combining Surfer Models with Operational Geometry and Volume Calculation (2007)

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Abstract

Modelor is a two-dimensional resource modeling software developed in 2007 for tabular (layered) ore deposits. The software was designed to construct grid-based models from exploration borehole data, generate contour maps and three-dimensional surfaces, create cross-sections in arbitrary directions, and calculate layer volumes. It employed the Inverse Distance to a Power interpolation method and the trapezoidal rule for volume estimation. Modelor was implemented in Delphi and used OpenGL for 3D visualization.

In practical application, the software was used to combine topography and pit models previously generated with Kriging in Surfer, incorporate operational geometry such as benches and ramps, and compute excavation volumes for cost estimation purposes. This paper describes the software architecture, workflow, real-world usage, validation against Surfer, and its inherent limitations. Modelor represents an early domain-oriented modeling effort that later informed operational systems in the oil and gas industry and eventually evolved into modern applied AI projects focused on automated cost and structural estimation.

Keywords: 2D resource modeling, tabular deposits, volume calculation, Inverse Distance, mining software, excavation cost estimation, Surfer integration

1. Introduction

Accurate resource modeling is a fundamental prerequisite for mine planning and economic evaluation. The choice between two-dimensional and three-dimensional modeling techniques depends largely on the geometry of the orebody. Tabular or layered deposits are typically better represented by 2D grid models, while massive or irregular deposits require 3D block models.

In the mid-2000s, commercial software packages such as Surfer (for 2D modeling) and Datamine (for 3D modeling) were the dominant tools available to mining engineers in Iran. However, no independent, locally developed software existed that specifically addressed the needs of 2D modeling for tabular deposits while remaining accessible and lightweight.

Modelor was developed in 2007 as one of the first attempts to fill this gap. The primary objectives were to provide a practical tool for constructing 2D grid models from exploration data, visualizing results through contours and 3D surfaces, generating cross-sections, and calculating volumes of layered units.

This paper presents the design, capabilities, practical application, and limitations of Modelor, and situates it within the author's longer professional trajectory in operational modeling and applied industrial AI.

2. Software Architecture and Workflow

The overall workflow of Modelor can be divided into three sequential stages.

2.1 Data Input and Preparation

Modelor accepts exploration data in a structure similar to that used by Datamine:

Collar file: Borehole identifiers and collar coordinates (X, Y, Z).

Assay file: Borehole identifiers, from-to depths, geological interpretation (layer codes), and grade values.

The software also includes a built-in digitizer that allows users to convert paper maps or topographic plans into digital point data. After importing the Collar and Assay files, the system merges them into a unified internal dataset ready for modeling.

2.2 Model Construction

The core modeling engine uses the Inverse Distance to a Power (IDW) interpolation method. Users define the modeling extent (minimum and maximum X and Y), grid cell size, the power parameter and smoothing factor, and the target variable (thickness, grade, elevation of roof or floor, etc.). The software then generates a regular 2D grid and estimates values at each grid node.

2.3 Analysis and Output Modules

Once a grid model is created, the following operations are available:

- Volume calculation using the trapezoidal method between two surfaces (e.g., roof and floor of a layer).
- Contour mapping with user-defined contour intervals and colors.
- 3D surface visualization via OpenGL, with the ability to overlay multiple surfaces and display borehole traces.
- Cross-section generation along straight or polyline paths.
- Tabular display of grid values.

3. Real-World Application: Combining Surfer Models with Operational Geometry

A representative practical application involved a layered copper deposit. Topography of the ground surface and the designed pit surfaces were first modeled using ordinary Kriging in Surfer. These surfaces were then imported into Modelor.

Within Modelor, operational design elements—specifically benches and ramps—were incorporated into the combined model. The software was subsequently used to calculate the volume of material to be excavated. This volume formed the quantitative basis for estimating extraction costs.

This workflow demonstrated two important capabilities of Modelor:

1. Interoperability with externally generated surfaces (from Surfer),
2. Support for cost-oriented decision-making through accurate volume computation that included operational geometry.

4. Validation

To assess the reliability of the results, identical input data and grid parameters were used in both Modelor and Surfer. Comparisons were performed on contour maps of the same variable, numerical values extracted along the same cross-section line, and volume calculations between the same pair of surfaces.

The results showed close agreement. Volume estimates produced by Modelor's trapezoidal method matched those obtained from Surfer under equivalent conditions.

5. Limitations

Despite its practical utility at the time, Modelor had several inherent limitations:

- Only a single interpolation technique (Inverse Distance to a Power) was implemented; more advanced geostatistical methods such as Kriging were not available.
- Boreholes were assumed to be vertical.
- Geological structures such as faults could not be modeled.
- The software was developed with 2007-era technology (Delphi and OpenGL).
- The source code is no longer available.

These constraints reflect both the scope of a bachelor's-level project and the technological context of the period.

6. Conclusion and Subsequent Evolution

Modelor provided early hands-on experience in spatial data modeling, grid-based interpolation, volume calculation, and the integration of geological and operational geometry. Although limited by the methods and technology of its time, the domain knowledge acquired during its development proved foundational.

In subsequent years, this experience contributed to the design of large-scale operational data systems in the oil and gas industry (including fleet tracking and drilling report platforms). More recently, the same underlying focus on geometry, volume, and cost estimation has been carried forward into modern applied AI projects such as Plan2Cost, which combines classical computer vision with hybrid machine-learning cost engines for automated estimation from floor plans.

Modelor therefore stands as an early milestone in a longer trajectory of domain-driven modeling and decision-support software.

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